MANIPAL INSTITUTE OF TECHNOLOGY

CES 2103 Data Analytics | IA3 Mini Project

Estimated and Recorded Smartphone Screen Time Among BTech Students

Team 3

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1 Introduction and Objective

How accurately do B.Tech students estimate their daily smartphone use? We study this question by comparing an estimate with the average of seven daily phone records. Both measurements refer to **the same person, primary phone and seven completed days**. This keeps the comparison meaningful.

Our objective is to measure the direction and size of estimation errors, then test whether the average difference is zero. Exploratory data analysis (EDA) describes the patterns. Confirmatory data analysis (CDA) tests the specific hypothesis using a paired t-test.

2 Dataset Source and Description

The current source is **data/simulated/screen_time_simulated.csv**, created in Python with NumPy and the fixed random seed 20261017. It contains **120 simulated records**, of which **112 pass validation**. These are generated observations; no Google Forms responses were used for this analysis.

Each row represents one simulated student with an estimate and seven daily totals. The generator includes different usage levels, day-to-day variation, a small weekend increase and estimates rounded to 15 minutes. It assumes mean underestimation of 20 minutes, so the resulting bias partly reflects the model design. Eight invalid records were added to exercise the cleaning rules.

The planned survey uses the same fields: respondents estimate first, then enter daily totals for the same dates. Names, college identifiers and phone operating systems are not required. The current analysis demonstrates the method; conclusions about actual students require genuine responses.

3 Data Preprocessing

We preserve the source CSV and create a separate cleaned table. **One row is one observation**: seven daily readings are averaged within that row rather than counted as seven independent students.

3.1 Variables and units

Table 1 Main variables used in the analysis

Variable	Meaning	Unit or type
Participant ID	Label assigned after import	Nominal
Estimated daily time	Estimate before checking records	Minutes per day
Seven daily totals	Usage for each completed day	Minutes per day
Recorded daily average	Mean of all seven daily totals	Minutes per day
Signed error	Recorded average minus estimate	Minutes per day
Absolute error	Magnitude of the signed error	Minutes per day

Durations arrive as H:MM text and are converted to minutes. For example, **4:30 = 270 minutes**, not 4.30 hours. The reference end date identifies the last day of the seven-day window. Five Yes/No fields record consent, B.Tech eligibility, first response, estimate-before-check confirmation and recording-scope confirmation.

3.2 Validation and exclusions

A row is included only when all five confirmations are Yes, the date is valid, and the estimate plus every daily reading is present and between 0:00 and 24:00. Eight rows fail: one has a missing daily total, one an impossible duration, one an invalid date, and five each fail a different confirmation. Thus **120 − 8 = 112 complete pairs** remain.

We do not replace missing durations with zero or an average. An incomplete week is excluded. Identical response patterns are flagged for review rather than automatically deleted, because matching values do not prove that the same person responded twice. The audit records each inclusion decision and its reason.

3.3 Calculated measures

$$\text{Recorded daily average} = \text{sum of the seven daily totals} / 7$$

$$\text{Signed error} = \text{recorded daily average} - \text{estimated daily time}$$

Positive error means underestimation; negative error means overestimation. Absolute error is the size of this difference without its sign. For example, an estimate of 240 minutes and a recorded average of 300 minutes gives a signed error of +60 minutes and an absolute error of 60 minutes.

4 Exploratory Data Analysis

EDA describes the 112 included pairs before the hypothesis test. The mean gives the average, the median is the middle value, and the sample standard deviation (SD) describes variation between records. All durations below are in minutes per day.

Table 2 Descriptive statistics for the included records

Measure	Mean	Median	SD
Estimated time	269.20	255.00	133.42
Recorded average	291.68	283.93	112.84
Signed error	22.48	28.00	72.17
Absolute error	59.42	45.57	46.44

4.1 Comparing estimates with recorded averages

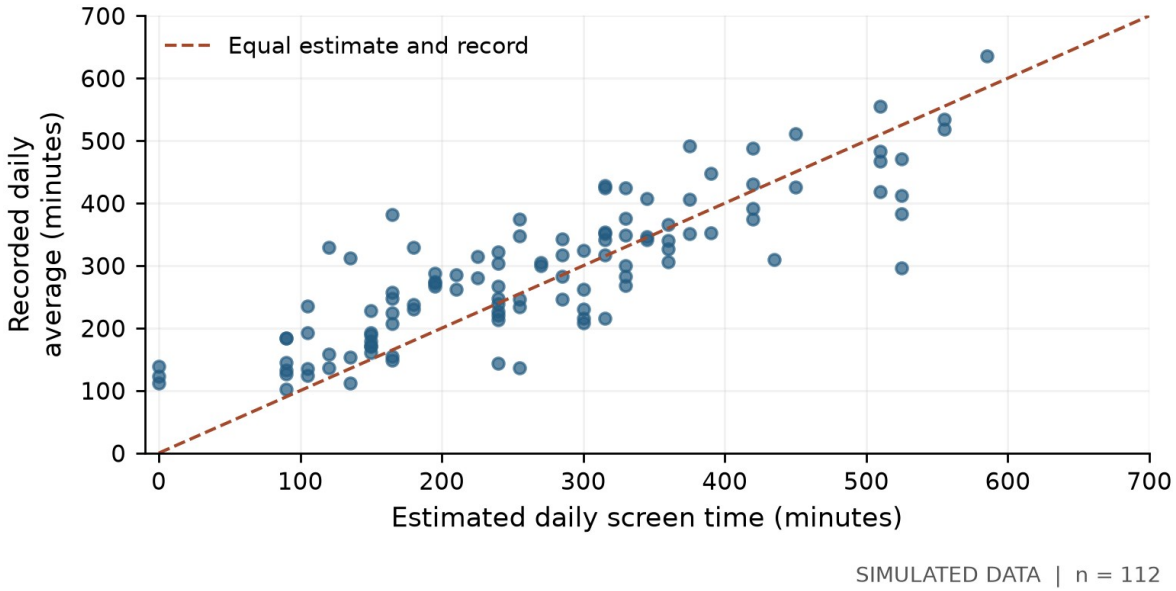


Figure 1 Each point represents one simulated record. The dashed line marks equal values.

Points above the dashed line show underestimation: the recorded average exceeds the estimate. Points below it show overestimation. Points close to the line have smaller errors. Although higher estimates generally accompany higher recorded use, the spread around the line shows that this association does not guarantee accurate estimates.

The mean estimate is **269.20 minutes**, while the recorded mean is **291.68 minutes**. Their difference is **22.48 minutes per day**. The recorded median is 283.93 minutes, slightly below its mean, consistent with some higher-usage records pulling the mean upward.

These summaries answer different questions. The two usage means describe overall levels, while the signed and absolute errors describe how closely each estimate matches its own recorded average.

4 Exploratory Data Analysis Continued

4.2 Direction and size of the errors

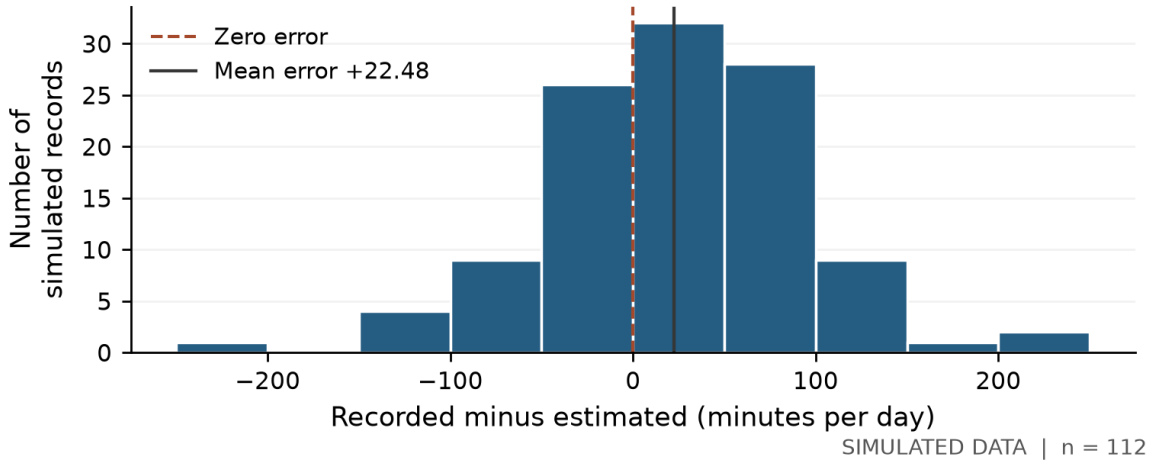


Figure 2 Errors occur on both sides of zero; the mean is positive.

Mean signed error is 22.48 minutes; mean absolute error is 59.42 minutes. Overestimates and underestimates partly cancel when we calculate the mean signed error. For example, errors of +60 and −60 average to zero, although both estimates are wrong by 60 minutes.

Using a descriptive 30-minute tolerance, 53 records (47.3%) underestimate by more than 30 minutes, 22 (19.6%) overestimate by more than 30 minutes, and 37 (33.0%) fall within 30 minutes. This tolerance is a project convention, not a health threshold or a validated accuracy standard.

4.3 Checking the differences before the test

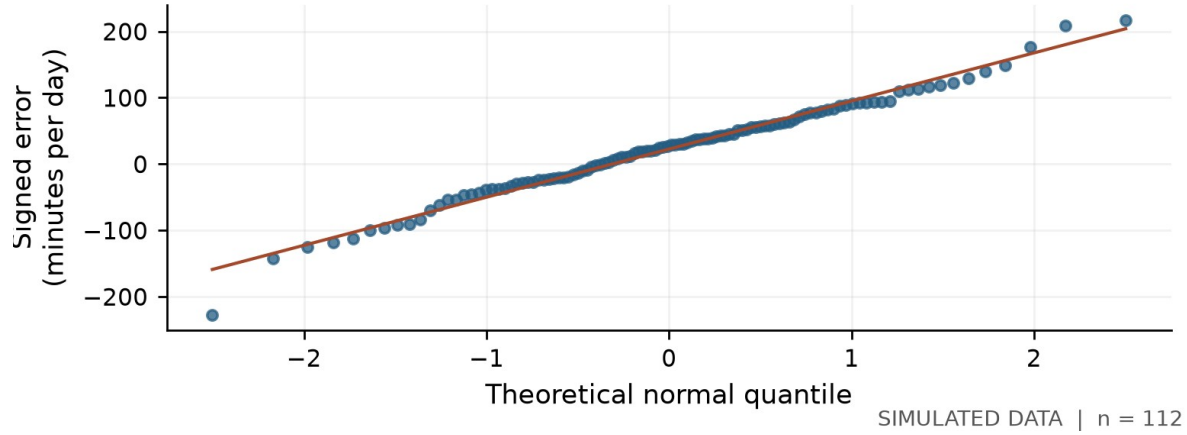


Figure 3 A normal Q–Q plot compares the observed differences with normal reference quantiles.

The Q–Q plot is roughly linear in the centre, with departures in the tails. Three differences fall outside the 1.5 × IQR fences, where IQR is the spread of the middle half of the data. We retain these valid values and report the departures. The paired t-test concerns the distribution of differences; these checks do not prove normality or independence.

5 Confirmatory Data Analysis

5.1 Hypothesis and test selection

We use a **two-sided paired t-test at α = 0.05**. Each estimate is matched to the same record's seven-day average, so the analysis tests the within-record differences [1]. The two-sided choice allows either underestimation or overestimation.

H_0: the population mean of recorded minus estimated time is zero. **H_1:** this mean difference is not zero. In this report, the population is the hypothetical one represented by the simulation.

The test assumes independent participant differences and a sufficiently well-behaved distribution of differences. The previous page shows tail departures to consider when interpreting the result. A large sample cannot correct biased sampling or inaccurate entries.

5.2 Calculation and result

The standard error (SE) describes uncertainty in the estimated mean. It differs from the SD, which describes variation across individual records. The test divides the mean difference by its SE.

$$SE = 72.1744 / \sqrt{112} = 6.8198 \text{ minutes}$$

$$t = 22.4847 / 6.8198 = 3.2970 \qquad df = 112 - 1 = 111$$

Table 3 Paired t-test results for the simulated dataset

Result	Value
Complete pairs	112
Mean difference	22.48 minutes per day
Test statistic and degrees of freedom	t = 3.297; df = 111
Two-sided p-value	0.00131
95% confidence interval	8.97 to 36.00 minutes per day
Standardised paired effect size	Cohen's dz = 0.312

5.3 Interpretation

Because **p = 0.00131 < 0.05**, we reject zero mean difference for this simulated analysis. If H_0 and the test assumptions held, a t-statistic at least this far from zero would be unlikely. The p-value is not the probability that H_0 is true.

The **95% confidence interval is 8.97 to 36.00 minutes per day** and excludes zero. It describes uncertainty in the population mean difference, not a range containing 95% of individual errors. Under repeated sampling with the model assumptions, 95% of intervals constructed this way would cover that mean.

Cohen's dz divides the mean difference by the SD of the differences: 22.48 / 72.17 = 0.312. The average bias is about 0.31 SD. The error in minutes remains the more direct measure of practical size.

6 Python Implementation and Reproducibility

The workflow is **load, validate, calculate errors, explore, test and export**. The notebook explains each step; reusable Python functions perform the calculations.

6.1 Libraries and core calculations

pandas organises the table; NumPy supports calculations. Matplotlib and Seaborn create plots, and SciPy provides the statistical test. CDA is the hypothesis-testing stage, rather than a separate Python library.

Here, numeric_days contains the seven duration columns. axis=1 calculates across each row; skipna=False prevents an incomplete week from being silently averaged.

```
frame["recorded_daily_minutes"] = frame[numeric_days].mean(
    axis=1, skipna=False
)
frame["signed_error_minutes"] = (
    frame["recorded_daily_minutes"]
    - frame["estimated_daily_minutes"]
)
```

SciPy compares the matched columns [2]. Their order makes underestimation positive. nan_policy="raise" stops the test if missing values remain.

```
test = stats.ttest_rel(
    clean["recorded_daily_minutes"],
    clean["estimated_daily_minutes"],
    alternative="two-sided", nan_policy="raise"
)
```

The code checks that the formula-based t and p-value match SciPy, and exports the confidence interval, effect size and outlier flags. Automated checks cover time conversion, exclusions, source handling and statistical calculations.

6.2 Code files and saved outputs

File or folder	Role
run.py	Command-line entry point
src/screentime/pipeline.py	Validation, EDA, paired test and exports
src/screentime/simulate.py	Reproducible simulation generator
notebooks/01_screen_time_analysis.ipynb	Explained analysis with outputs
config/ and outputs/	Input settings and saved analysis results

Run python run.py --config config/simulated.json. Outputs include cleaned data, an audit, descriptive statistics, figures and statistics.json. A manifest records the source checksum, settings and software versions.

For a real survey, add data/raw/screen_time_real.csv, map form headings in config/real.json and rerun with that configuration. Refresh the notebook and report from the new outputs. Missing real data never trigger a fallback to simulation.

7 Findings and Conclusions

7.1 What the analysis shows

The generated records show **average underestimation of 22.48 minutes per day**, but an **average absolute error of 59.42 minutes**. This distinction is the main finding: a modest net bias can hide much larger errors in individual estimates.

EDA shows errors in both directions and wide variation between records. CDA finds evidence of a nonzero mean difference within the simulated model. These results answer complementary questions: the paired test addresses average bias, while the plots and absolute errors describe the spread and size of the mismatches.

7.2 Limitations and next step

The main limitation is the source. The data are simulated, and the generator already assumes some underestimation. The significant result therefore partly reflects those assumptions; it is not independent evidence about B.Tech students. The current project establishes a working analysis that can be applied to collected responses.

A future survey shared through friends would be a convenience sample, so it would not represent all B.Tech students. Respondents might mistype device totals or change an estimate after checking. An anonymous form also cannot fully verify unique participation. Keeping the same phone, dates and instructions helps consistency but does not remove these limitations.

The next step is to pilot the form, collect genuine responses and rerun the same analysis. The new results may support a different conclusion. Simulated and real records should remain separate, and every reported number and chart must be updated after a change of source.

7.3 Conclusion

We implemented the full process from data validation to statistical interpretation. The analysis explains why mean bias alone is not enough to judge estimation accuracy: the direction, absolute size and spread of the errors all matter. With genuine survey data, this workflow can assess how closely respondents' estimates match their phone records.

Project Repository

https://github.com/itsmePrithviraj/ia3-data-analytics

References

[1] NIST/SEMATECH. e-Handbook of Statistical Methods, Two-Sample t-Test for Equal Means, discussion of paired observations. Accessed 26 October 2026.

https://www.itl.nist.gov/div898/handbook/eda/section3/eda353.htm

[2] SciPy documentation. scipy.stats.ttest_rel, related-sample t-test API. Accessed 26 October 2026.

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_rel.html